

Spark Homes Repair Estimator

Developer Contest Writeup
Submitted July 2026

1. MOST INTERESTING DESIGN DECISION

The room tab bar uses swipe gestures to navigate between rooms. Agents in the field hold their phone with one hand while taking notes or photos with the other. Tapping small tabs repeatedly on a phone screen is slow and error-prone in that context. Left and right swipes let agents move through the property room by room without repositioning their grip. The Vibration API fires a short haptic pulse on each swipe so the transition registers without the agent needing to look at the screen.

The second decision was treating the Deal Analyzer as a first-class feature rather than a setting. The repair total is only useful if the agent can immediately answer whether the deal still works at the asking price. The Deal button sits in the header on every screen so that number is always one tap away.

2. WHAT IS BROKEN OR FRAGILE

Area	Issue	Fix
OCR	Requires network. Fails silently offline.	Cache model or use on-device ML
Photos	Base64 in localStorage hits quota above 5-10 photos	Move to IndexedDB
Offline CDN	Tailwind, XLSX, JSZip need first-load network	Self-host or inline all scripts
Multi-device	Two agents on same project causes data collision	Add a lightweight backend

3. CREATIVE ADDITION: DEAL ANALYZER

The brief asks for a repair estimator. But the actual job agents need done at the end of a walkthrough is deciding whether to make an offer. The repair total is one input into that decision, not the output.

The Deal Analyzer takes ARV, purchase price, and holding costs alongside the live repair total and shows: projected profit, ROI, deal health rating (color coded by margin), and Max Allowable Offer using the 70 percent rule ($ARV \times 0.70$ minus repairs minus holding costs). If the purchase price already exceeds the MAO, a warning appears immediately. A share button copies the full deal snapshot as plain text for texting to the acquisition manager.

This turns a cost-tracking tool into a deal decision tool, which is the real job.

4. WHAT I WOULD SHIP NEXT WITH TWO MORE DAYS

IndexedDB for photos. Replace localStorage base64 with IndexedDB blobs to remove the storage ceiling.

Shareable project URL. Encode project state into a compressed URL hash so agents can hand off estimates without emailing files.

Offline CDN bundle. Inline Tailwind, XLSX, and JSZip into the HTML so the app works fully without any network connection.

Contractor quick-dial. Per-trade contact list so agents can tap a category and immediately call the right contractor.

AI TOOLS

This app was built using Claude by Anthropic. Claude wrote the implementation based on the brief and price list. Architecture decisions, UX direction, and feature choices were directed and iterated on by the submitter. Claude also powers the serial number OCR feature at runtime via the Vision API.